ECE 285 Final Project Report: *SolarInvest Agent*

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Abstract

This project proposes SolarInvestAgent, a novel intelligent decision-support system that optimizes the sizing of residential Photovoltaic (PV) systems and Battery Energy Storage Systems (BESS). Unlike standard Large Language Models (LLMs) that rely primarily on heuristic reasoning, the system integrates a rigorous Mixed-Integer Nonlinear Programming (MINLP) formulation to perform constrained optimization. Following the methodology established by Zhou et al. (2018), the agent explicitly models tariff structures (Stepwise and Time-of-Use), seasonal weather variability, and battery degradation constraints to produce mathematically grounded investment recommendations. The system was evaluated using five years of historical household consumption data to capture demand variability and determine the optimal PV technology, system capacity, and installation size that minimize total lifecycle cost under tariff, weather, and degradation constraints. [Include one more line about tools]

1 Introduction

Residential PV adoption is increasing, but selecting the right PV and battery size remains difficult under Time-of-Use tariffs, climate variability, and battery aging. Existing decision workflows are either technically rigorous but hard for non-experts, or easy to access (LLM-only) but often heuristic and unreliable.

We formulate residential PV-BESS sizing as a constrained optimization problem: given weather, load, tariff, and technology-cost inputs, choose PV and battery capacities that minimize lifecycle electricity cost while satisfying power-balance and battery-operation constraints. SolarInvestAgent combines an LLM interface for orchestration and explanation with a solver-backed optimization core. We compare it against an LLM-only baseline across household/location scenarios using financial accuracy, tariff sensitivity, and feasibility metrics. The optimization-grounded agent yields more consistent, constraint-satisfying recommendations.

We developed SolarInvest Agent, a hybrid decision-support architecture that integrates mathematical optimization with an LLM interface. The system couples a PV-BESS sizing optimizer with a data pipeline that extracts weather, tariff, and load information for each location, enabling mathematically grounded recommendations through a natural-language interface.

We evaluate the proposed system on a custom benchmark spanning extreme operating scenarios to assess reasoning quality and response accuracy. We perform inference with both a baseline LLM and SolarInvest Agent, and compare outputs against manual evaluation to assess correctness.

Our results show that embedding constrained optimization within an agentic AI framework produces more reliable recommendations than a heuristic LLM baseline. The optimization-based approach consistently satisfies physical battery constraints and adapts to Time-of-Use pricing, while the LLM-only baseline frequently generates economically inconsistent or infeasible configurations.

Preprint.

2 Related Work

The problem of optimal sizing of photovoltaic and battery energy storage systems has been widely studied within the smart-grid and energy management literature. Early foundational work by Zhou et al. formulated the joint PV and BESS sizing problem as a mixed-integer nonlinear programming (MINLP) optimization under multiple electricity pricing mechanisms, demonstrating that tariff structure significantly influences optimal system design Zhou et al. [2018]. Their results showed that PV-only installations may be optimal under stepwise tariffs, whereas combined PV–BESS configurations become economically attractive under Time-of-Use pricing due to energy arbitrage opportunities.

Subsequent research has extended these formulations through stochastic optimization, mixed-integer linear programming, and scenario-based modeling approaches that explicitly account for uncertainty in solar generation and electricity demand. These studies incorporate additional considerations such as battery degradation, demand response participation, and variability in household consumption patterns, all of which influence optimal storage sizing and operational strategies Afzal et al. [2018]; Hanna et al. [2017]; Wang et al. [2024]. Collectively, this body of work establishes that accurate modeling of tariffs, load variability, and solar generation is critical for determining economically viable residential solar installations.

More recently, artificial intelligence methods have been explored as decision-support tools in energy systems. Large language models and generative AI techniques have been applied to tasks such as energy forecasting, operational coordination, and policy analysis in smart grids Bahi et al. [2026]. However, these models typically operate as advisory systems rather than physically grounded optimization tools. Their recommendations may therefore lack quantitative guarantees regarding feasibility or cost optimality.

This project builds on the PV–BESS sizing literature by embedding an optimization-based energy system model within an LLM-driven interface. Instead of replacing optimization with language models, the proposed architecture uses the LLM as an orchestration and explanation layer while delegating numerical decision-making to a constrained solver. This hybrid approach bridges the gap between mathematically rigorous energy system design methods and user-friendly conversational interfaces for residential energy planning.

3 Problem Formulation

We model a grid-connected residential energy system consisting of a photovoltaic (PV) array, a battery energy storage system (BESS), and grid interaction. The system operates over discrete time steps indexed by $t \in \{1, \ldots, T\}$, where $T = 8760$ represents hourly resolution over one year. For stochastic extensions, scenarios are indexed by $s \in \mathcal{S}$.

Symbol	Definition
S_{PV}	Installed PV capacity (kWp)
E_{BESS}	Battery energy capacity (kWh)
$P_{\text{buy}}^{\text{grid}}(t),\ P_{\text{sell}}^{\text{grid}}(t)$	Grid import/export power (kW)
$P_{\text{ch}}(t),\ P_{\text{dch}}(t)$	Battery charge/discharge power (kW)
$\text{SoC}(t)$	Battery state-of-charge (kWh)
$P_{\text{load}}(t)$	Household demand
$P_{\text{PV}}(t)$	PV generation
$\lambda_{\text{TOU}}^{\text{buy}}(t),\ \lambda_{\text{NEM}}^{\text{sell}}(t)$	Time-varying buy/sell tariffs

Table 1: Compact notation summary for decision variables and exogenous inputs.

The objective is to minimize the total electricity cost over the planning horizon:

$$\min C_{\text{total}} = C_{\text{PV}} + C_{\text{BESS}} + C_{\text{O\&M}} + \sum_{t=1}^{T} \left[P_{\text{buy}}^{\text{grid}}(t) \cdot \lambda_{\text{TOU}}^{\text{buy}}(t) - P_{\text{sell}}^{\text{grid}}(t) \cdot \lambda_{\text{NEM}}^{\text{sell}}(t) \right] \tag{1}$$

subject to the following constraints:

(1) Power Balance

$$P_{\text{load}}(t) = P_{\text{buy}}^{\text{grid}}(t) - P_{\text{sell}}^{\text{grid}}(t) + P_{\text{PV}}(t) + P_{\text{dch}}(t) - P_{\text{ch}}(t) \tag{2}$$

(2) Battery SoC Dynamics

$$\text{SoC}(t) = \text{SoC}(t-1) + \eta_{\text{ch}} P_{\text{ch}}(t) - \frac{P_{\text{dch}}(t)}{\eta_{\text{dch}}} \tag{3}$$

(3) SoC Bounds

$$0.2 \cdot E_{\text{BESS}} \leq \text{SoC}(t) \leq 0.8 \cdot E_{\text{BESS}} \tag{4}$$

(4) Charge/Discharge Exclusivity

$$x_{\text{ch}}(t) + x_{\text{dch}}(t) \leq 1, \quad x_{\text{ch}}, x_{\text{dch}} \in \{0, 1\} \tag{5}$$

(5) PV Generation Constraint

$$P_{\text{PV}}(t) = S_{\text{PV}} \cdot \frac{G(t)}{G_{\text{ref}}} \tag{6}$$

Irradiance Model. Sunset hour for day d:

$$S(d) = 18 + 2\sin\left(\tfrac{2\pi(d-80)}{365}\right), \quad r = 6 \text{ (sunrise)} \tag{7}$$

Normalised hourly shape factor:

$$\phi(h, d) = \max\left(0,\ \sin\left(\frac{\pi(h-r)}{S(d)-r}\right)\right) \cdot \mathbf{1}[r \leq h < S(d)] \tag{8}$$

Calibrated hourly irradiance ($\eta_{\text{PR}} = 0.80$, G = annual kWh m^{-2} yr^{-1}):

$$G_t = \phi(h_t, d_t) \cdot \frac{G\,\eta_{\text{PR}}}{\sum_{t'} \phi(h_{t'}, d_{t'})} \tag{9}$$

API conversion from daily shortwave radiation $\bar{R}_d$ [MJ m^{-2} day^{-1}]:

$$G = \frac{\bar{R}_d}{3.6} \times 365 \quad [\text{kWh m}^{-2}\text{yr}^{-1}] \tag{10}$$

PV System Hourly output for p panels of rated power P_{W} [W]:

$$P_t^{\text{PV}} = \frac{p\,P_{\text{W}}}{1000} \cdot G_t \tag{11}$$

Annual production and per-panel yield:

$$E^{\text{PV}} = \frac{p\,P_{\text{W}}}{1000} \cdot G\,\eta_{\text{PR}}, \qquad E_{\text{panel}} = \frac{P_{\text{W}}}{1000} \cdot G\,\eta_{\text{PR}} \tag{12}$$

Panel selection (no brand specified): $\text{panel}^* = \arg\max_j \frac{\eta_j}{c_j}$

Load Profile Let $\Delta\lambda = \lambda - \lambda_{\text{coast}}$ ($\lambda_{\text{coast}} = -117.25$), φ = latitude, $d_c = \sqrt{(\varphi - 32.7157)^2 + (\lambda + 117.1611)^2}$.

$$f_1 = \begin{cases} 1.25 & \Delta\lambda \geq 0.15 \\ 1.05 + 4(\Delta\lambda - 0.10) & 0.10 \leq \Delta\lambda < 0.15 \\ 0.95 + \Delta\lambda & 0 \leq \Delta\lambda < 0.10 \\ 0.90 + (\Delta\lambda + 0.05) & -0.05 \leq \Delta\lambda < 0 \\ 0.85 & \Delta\lambda < -0.05 \end{cases} \qquad f_2 = \begin{cases} 1.10 & \varphi < 32.60 \\ 1.05 & 32.60 \leq \varphi < 32.70 \\ 1.00 & 32.70 \leq \varphi < 32.85 \\ 0.95 & 32.85 \leq \varphi < 32.95 \\ 0.90 & \varphi \geq 32.95 \end{cases} \tag{13}$$

$$f_3 = 1 + 0.15\big[\max(0, \Delta\lambda) + 2\max(0, \varphi - 32.70)\big], \quad f_4 = 0.70 + 0.10\min(n_p, 6) \tag{14}$$

$$f_5 = \begin{cases} 0.7 & d_c < 0.03 \\ 0.9 & 0.03 \leq d_c < 0.08 \\ 1.1 & 0.08 \leq d_c < 0.15 \\ 1.3 & d_c \geq 0.15 \end{cases} \tag{15}$$

Scaled load profile (f_6: coastal/urban factor; f_9: stochastic micro-grid factor):

$$L_t = L_t^{\text{base}} \cdot f_1 f_2 f_3 f_4 f_5 f_6 f_9 \tag{16}$$

EV charging overlay ($P_{\text{EV}} = 7.2$ kW, $E_{\text{daily}} = 14.0$ kWh EV^{-1} day^{-1}, hours 22:00-06:00):

$$L_t^{\text{EV}} = P_{\text{EV}}\, n_{\text{EV}} \cdot \mathbf{1}[t \bmod 24 \in \mathcal{H}_{\text{charge}}], \quad |\mathcal{H}_{\text{charge}}| = \left\lceil \frac{E_{\text{daily}}}{P_{\text{EV}}} \right\rceil \tag{17}$$

Daytime occupancy uplift ($9 \leq h < 17$):

$$L_t \leftarrow L_t \cdot (1 + 0.05\, n_{\text{day}}) \tag{18}$$

Battery Dispatch Simulation Net at hour t: $\text{net}_t = L_t - P_t^{\text{PV}}$

Surplus ($\text{net}_t < 0$):

$$c_t = \min\left(P_c^{\max},\ \frac{C - \text{SoC}_t}{\eta},\ |\text{net}_t|\right), \quad e_t = |\text{net}_t| - c_t \tag{19}$$

Deficit ($\text{net}_t \geq 0$):

$$d_t = \min(P_d^{\max},\ \text{SoC}_t,\ \text{net}_t), \quad g_t = \text{net}_t - d_t \tag{20}$$

SoC update:

$$\text{SoC}_{t+1} = \text{clip}(\text{SoC}_t + c_t\,\eta - d_t,\ 0,\ C) \tag{21}$$

Annual cost terms:

$$\text{CostImport} = \sum_t g_t\,\pi_t, \qquad \text{CreditExport} = \sum_t e_t\,\pi_{\text{NEM}} \tag{22}$$

Economic Model Capital costs ($r_{\text{inst}} = 0.10$, $r_{\text{ITC}} = 0.30$):

$$\text{CAPEX}_{\text{gross}} = \big(p\,P_{\text{W}}\,c_{\text{W}} + n_{\text{units}}\,c_{\text{bat}}\big)(1 + r_{\text{inst}}), \quad \text{CAPEX}_{\text{net}} = \text{CAPEX}_{\text{gross}}(1 - r_{\text{ITC}}) \tag{23}$$

Annual bills and year-1 savings ($F = \$0.345\,\text{day}^{-1}$, $c_{\text{O\&M}} = \$0.005\,\text{W}^{-1}\,\text{yr}^{-1}$):

$$B_1^{\text{trad}} = E^{\text{load}}\,\bar{\pi} + 365F \tag{24}$$

$$B_1^{\text{solar}} = \text{CostImport} - \text{CreditExport} + 365F \tag{25}$$

$$S_1 = B_1^{\text{trad}} - B_1^{\text{solar}} - c_{\text{O\&M}}\,p\,P_{\text{W}} \tag{26}$$

Multi-year savings with degradation rate δ and utility inflation $r_{\text{inf}} = 0.06$:

$$E_y^{\text{imp}} = E^{\text{load}} - \big(E^{\text{load}} - E_{\text{sim}}^{\text{imp}}\big)(1 - \delta)^{y-1} \tag{27}$$

$$S_y = B_1^{\text{trad}}(1 + r_{\text{inf}})^{y-1} - \Big[E_y^{\text{imp}}\,\bar{\pi}_{\text{eff}}(1 + r_{\text{inf}})^{y-1} - E_y^{\text{exp}}\,\pi_{\text{NEM}} + 365F + c_{\text{O\&M}}\,p\,P_{\text{W}}(1 + r_{\text{inf}})^{y-1}\Big] - I_y \tag{28}$$

where $E_y^{\text{exp}} = E_{\text{sim}}^{\text{exp}}(1 - \delta)^{y-1}$ and $I_{10} = \$2{,}000$ (inverter replacement), $I_y = 0$ otherwise.

NPV and payback ($r_d = 0.07$, $N = 10\,\text{yr}$):

$$\text{NPV} = -\text{CAPEX}_{\text{net}} + \sum_{y=1}^{N} \frac{S_y}{(1 + r_d)^y}, \qquad y^* = \min\left\{y \;\middle|\; \sum_{i=1}^{y} S_i \geq \text{CAPEX}_{\text{net}}\right\} \tag{29}$$

Annuity factor and present-value multiplier (used in MILP objective):

$$\text{AF} = \frac{r_d(1 + r_d)^N}{(1 + r_d)^N - 1}, \qquad \text{PVM} = \sum_{y=1}^{N} \frac{(1 + r_{\text{inf}})^{y-1}}{(1 + r_d)^y} \tag{30}$$

MILP Formulation **Decision variables** $p \in \mathbb{Z}_{\geq 0}$ (panels); $y_k \in \{0, 1\}$ (battery option); $g_t, e_t, c_{kt}, d_{kt}, \text{SoC}_t \geq 0$.

Objective — minimise annualised life-cycle cost

$$\min \quad \underbrace{c_p^{\text{net}} \cdot p \cdot \text{AF}}_{\text{PV annuity}} + \underbrace{\sum_k c_k^{\text{bat,net}}\,y_k \cdot \text{AF}}_{\text{battery annuity}} + \underbrace{\text{PVM}\sum_t \pi_t\,g_t}_{\text{import cost}} - \underbrace{\pi_{\text{NEM}}\,\text{PVM}\sum_t e_t}_{\text{export credit}} \tag{31}$$

where $c_p^{\text{net}} = P_{\text{W}}\,c_{\text{W}}(1 + r_{\text{inst}})(1 - r_{\text{ITC}})$ and $c_k^{\text{bat,net}} = c_k^{\text{bat}}(1 + r_{\text{inst}})(1 - r_{\text{ITC}})$.

Constraints

Battery option selection:

$$\sum_k y_k = 1; \qquad C = \sum_k C_k\,y_k; \quad P^c = \sum_k P_k^{c,\max}\,y_k; \quad P^d = \sum_k P_k^{d,\max}\,y_k \tag{32}$$

Power balance $\forall\, t$:

$$\frac{p\,P_{\text{W}}}{1000}G_t + g_t - e_t + \sum_k (d_{kt} - c_{kt}) = L_t \tag{33}$$

SoC dynamics $\forall\, t$:

$$\mathrm{SoC}_{t+1} - \mathrm{SoC}_t + \sum_k \frac{d_{kt}}{\eta_k} - \sum_k \eta_k\, c_{kt} = 0 \tag{34}$$

SoC bounds and initialisation ($\alpha_{\text{min}} = 0.10$, $\alpha_{\text{init}} = 0.50$):

$$\mathrm{SoC}_0 = \alpha_{\text{init}}\, C; \qquad \alpha_{\text{min}}\, C \le \mathrm{SoC}_t \le C \quad \forall\, t \tag{35}$$

Big-M coupling (charge/discharge only when option k selected):

$$c_{kt} \le M\, y_k, \quad d_{kt} \le M\, y_k \quad \forall\, k, t \tag{36}$$

Export cap, roof area, and budget:

$$e_t \le \tfrac{p\, P_{\text{W}}}{1000} G_t \quad \forall\, t \tag{37}$$

$$p\, A_{\text{panel}} \le A_{\text{roof}}, \quad A_{\text{roof}} = \max\!\left(\left\lfloor \tfrac{W}{w} \right\rfloor \left\lfloor \tfrac{L}{\ell} \right\rfloor,\ \left\lfloor \tfrac{W}{\ell} \right\rfloor \left\lfloor \tfrac{L}{w} \right\rfloor\right) A_{\text{panel}} \tag{38}$$

$$p\, c_{\text{panel}}(1 + r_{\text{inst}}) + \sum_k c_k^{\text{bat}}(1 + r_{\text{inst}})\, y_k \le B \tag{39}$$

String Configuration Optimal series/parallel split minimising deviation from 400 V inverter target:

$$n_s^* = \arg \min_{n_s \mid n_s \cdot n_p = n} |n_s\, V_{\text{mp}} - 400|, \quad V_{\text{string}} = n_s^*\, V_{\text{mp}} \tag{40}$$

Cell-level quantities:

$$V_{\text{cell}} = \frac{V_{\text{mp}}}{N_s^{\text{cell}}}, \qquad I_{\text{cell}} = \frac{P_{\text{W}}}{N_s^{\text{cell}}\, N_p^{\text{cell}}\, V_{\text{cell}}} \tag{41}$$

To ensure robustness under variability in weather and demand, the formulation is extended over scenarios:

$$\min \sum_{s \in \mathcal{S}} \pi_s \cdot C_{\text{total},s}(S_{\text{PV}}, E_{\text{BESS}}) \tag{42}$$

where π_s represents the probability of each scenario.

The model outputs system-level recommendations as optimal sizing decisions $(S^*_{\text{PV}}, E^*_{\text{BESS}})$, with associated cost and savings estimates. In the agentic pipeline, outputs are structured into two scenario-level recommendations (*optimal* and *recommended*) plus a shared evidence block. Each scenario includes sizing parameters, energy estimates, financial metrics, constraints, assumptions, and confidence, yielding a machine-readable record. A predefined JSON schema specifies required fields, data types, and valid ranges to ensure consistency, interpretability, and downstream compatibility.

A recommendation is valid only if physical and logical constraints are satisfied, including feasible sizing, bounded operational states (e.g., $\mathrm{SoC}(t)$), and time-step power-balance consistency. Validation proceeds in three stages: syntactic (valid JSON), schema (structural correctness and bounds), and logical (domain checks such as budget feasibility and realistic configurations). If any check fails, an automatic repair step prompts the model to regenerate output under the same schema constraints. This validation-repair loop improves robustness while preserving evidence traceability to computed features and retrieved context.

4 Methodology

4.1 System Overview

We design a two-track architecture for residential PV system sizing: (i) a direct LLM-based reasoning pipeline, and (ii) an agentic optimization-driven system (SolarInvestAgent). Both tracks share a common data and feature processing backbone, but differ fundamentally in how decisions are generated. The LLM track treats the model as the primary reasoning engine that directly outputs recommendations, while the agentic track augments the reasoning process with deterministic mathematical tools and constraint-aware optimization.

This separation is motivated by a fundamental mismatch between generative reasoning and strict constraint satisfaction: while the LLM baseline produced coherent explanations, it often violated numerical and physical feasibility, and increasing prompt complexity led to context overload rather than improved reliability. We therefore redesigned the system as a hybrid agentic framework where deterministic tools handle constrained computation and the LLM synthesizes decisions, improving both correctness and flexibility.

4.2 Shared Data and Feature Pipeline

Both design tracks operate on a common preprocessing pipeline consisting of data extraction, stochastic household modeling, feature engineering, and retrieval augmentation.

4.2.1 Data Sources and Extraction

The system integrates three primary data streams: (i) five years of historical weather data (irradiance, temperature, cloud cover), (ii) regional electricity demand data, and (iii) derived household-level consumption profiles. These are temporally aligned and aggregated into weekly representations, forming a structured dataset: The full set of modeled San Diego County locations and their geospatial coordinates is listed in Appendix A.4 (Table 9).

$$\mathcal{D} = \{(w_t, l_t)\}_{t=1}^{T}, \quad T \approx 260$$

4.2.2 Stochastic Household Modeling

To bridge the gap between regional and household-level data, we construct a stochastic generative model that produces realistic consumption profiles using geographic and behavioral factors. The household load is modeled as:

$$P_t^{\text{house}} = P_t^{\text{base}} \cdot \prod_{i=1}^{k} f_i + g_{\text{EV}}(t) - g_{\text{solar}}(t)$$

where multiplicative factors capture climate, demographics, and infrastructure effects, and additive components model EV charging and solar offsets. Deterministic seeding ensures reproducibility across runs. Time-of-Use tariff assumptions used for pricing and sensitivity analysis are summarized in Appendix A.5 (Table 10).

4.2.3 Feature Engineering and Aggregation

Raw time-series data is compressed into a structured feature space consisting of over 75 domain-specific attributes spanning consumption patterns, solar potential, financial metrics, and risk indicators. In addition to baseline summary statistics, we explicitly compute sensitivity features to quantify economic robustness. For price sensitivity ($\delta = 0.10$), we evaluate `roi_under_price_change` using $\text{ROI}_{\text{baseline}} = \text{ROI}(p)$, $\text{ROI}_{\uparrow p} = \text{ROI}(p(1+\delta))$, and $\text{ROI}_{\downarrow p} = \text{ROI}(p(1-\delta))$, where

$$\text{ROI}(\%) = \frac{\text{Total_savings} - \text{Cost}}{\text{Cost}} \times 100, \quad \text{Total_savings} = \sum_{y=1}^{25} \left[s\,(1+e)^y \right], \quad s = \text{prod} \cdot p,$$

$$\text{prod} = \text{est._ann._production_per_panel} \times n, \quad \text{Cost} = \text{PV_INSTALL_FIXED_COST} + n \cdot \text{PV_PANEL_COST}.$$

We use defaults $p = 0.31$ USD/kWh, $e = 0.03$, and $\delta = 0.10$. For irradiance sensitivity (`roi_under_irradiance_change`), we compute $\text{ROI}_{\text{baseline}}$ on the original weather frame and recompute $\text{ROI}_{\uparrow I}$ and $\text{ROI}_{\downarrow I}$ after scaling weekly average irradiance by $(1 \pm \delta)$ in copied weather tables.

We also include volatility-based risk features. Consumption volatility is measured as coefficient of variation $\text{CV}_{\text{load}} = \sigma_{\text{load}}/\mu_{\text{load}}$ over weekly aggregated load (reported as `consumption_volatility_score`), and sunlight volatility as $\text{CV}_{\text{irr}} = \sigma_{\text{irr}}/\mu_{\text{irr}}$ over weekly irradiance (reported as `sunlight_volatility_score`). A combined risk score is then computed from these components in the feature-engineering stage and used downstream for uncertainty-aware recommendation ranking. The full workflow (including ROI sensitivity features and volatility-derived risk signals) is moved to Appendix A.1 (Figure 3).

4.2.4 Retrieval-Augmented Knowledge

Both pipelines incorporate a shared retrieval layer that provides domain grounding. Relevant knowledge about PV panel technologies, battery systems, tariffs, and installation practices is retrieved and injected into the reasoning context. This ensures that decisions are informed by both computed features and external domain knowledge. For panel hardware assumptions, we use market-referenced installed costs and efficiencies summarized in Appendix A.6 (Table 11). As representative endpoints, REC Group (Alpha Pure) provides a strong overall value point at \$2.85/W_p (22.6% efficiency), while Aiko Solar (Neostar 2P) represents the highest-efficiency premium option at 24.3% with higher installed cost; intermediate brand cases are documented in the appendix table.

4.3 LLM-Based Design (Baseline)

The first design treats the LLM as the primary reasoning engine. The model receives structured features, retrieved knowledge, and explicit constraints through a hierarchical prompt and directly produces PV sizing recommendations.

4.3.1 Prompt-Based Reasoning

The input is structured into three components: (i) feature-engineered data context, (ii) retrieved knowledge, and (iii) explicit task instructions with constraints. The instruction block follows an 8-step workflow: review data and knowledge jointly, identify trends/anomalies, account for EV charging demand, enforce budget feasibility, incorporate roof/occupancy/brand context, propose panel count with numeric justification, estimate savings and payback, and report risk factors with battery advisability. Constraint handling is explicit: hard constraints include budget, roof area, and EV load; soft constraints include panel-brand preference and daytime occupancy. The prompt also imposes an output schema requiring panel count (within budget), expected annual production, cost savings, payback period, ROI, EV charging impact, and caveats, so the model produces auditable, decision-ready recommendations rather than free-form narrative text.

4.3.2 Limitations of Direct LLM Inference

Despite incorporating structured prompts and explicit physical and financial constraints, the LLM-based approach exhibited several limitations. First, the model struggled to consistently satisfy hard constraints such as budget limits and roof area restrictions. Second, numerical reasoning involving financial metrics (e.g., ROI, payback) was often inconsistent, as the model approximated rather than strictly computed values. Third, increasing prompt complexity to enforce constraints led to degraded performance due to context saturation. Even with strong prompt engineering, the model lacks guarantees of feasibility, leading to outputs that may appear plausible but violate underlying constraints.

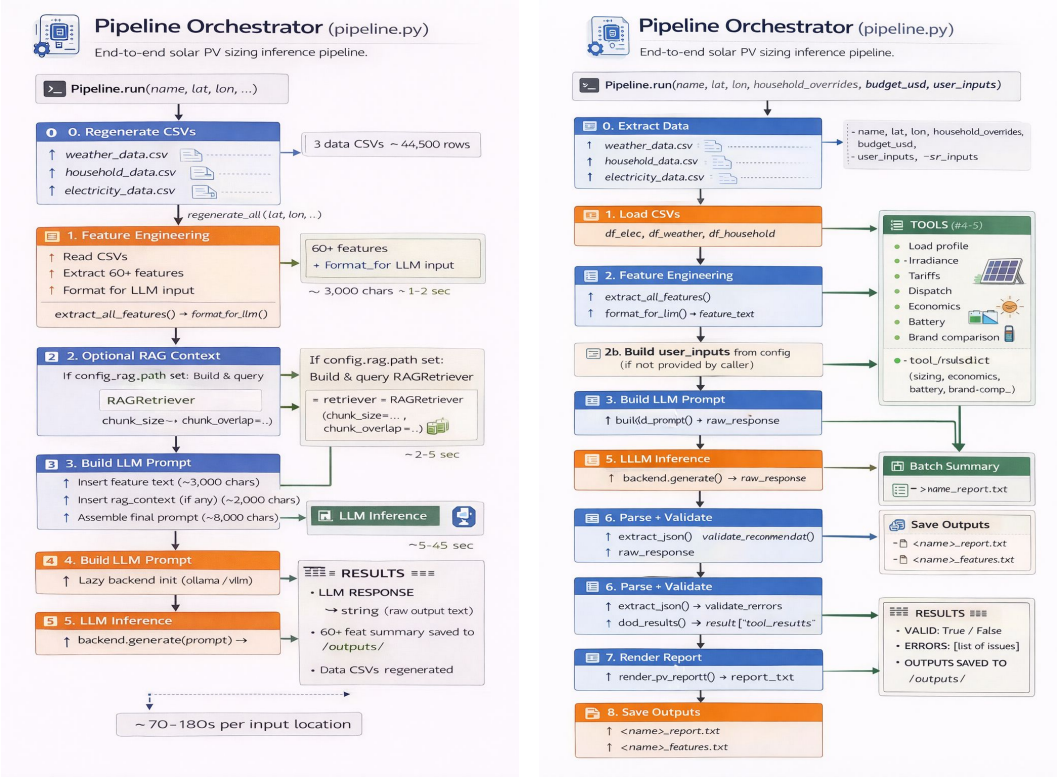

Figure 1: Track A (left) and Track B (right) pipeline overviews used in the system design.

4.4 Agentic Design: SolarInvestAgent

To address the limitations of direct LLM inference, we propose SolarInvestAgent, an agentic system that integrates deterministic mathematical computation with LLM-based reasoning. In this design, the LLM no longer directly produces final answers, but instead operates as a decision-making layer over a set of computational tools. Our proposed system extend LLMs by enabling them to plan, reason, and interact with external tools to achieve goal-oriented outcomes, rather than merely generating text. This shift allows the system to enforce correctness through computation while retaining flexibility in reasoning.

4.4.1 Tool-Augmented Mathematical Reasoning

All computations are implemented as deterministic functions grounded in physical and financial models. Photovoltaic generation is derived from irradiance-scaled panel output over a full 8760-hour horizon, while load and grid interactions are computed consistently in power and energy terms. This ensures that all quantities generation, consumption, and grid exchange follow explicit mathematical relationships.

System sizing and operation are jointly optimized using a mixed-integer linear program (MILP), where panel count, battery capacity, and hourly dispatch are solved together. The objective combines annualized capital cost with time-varying electricity costs and export credits. Financial metrics such as NPV and payback are then computed through deterministic cash-flow models, ensuring consistency between technical outputs and economic evaluation.

4.4.2 Constraint Enforcement

All feasibility conditions are enforced explicitly within the optimization layer rather than being left to interpretation. Physical constraints such as roof area directly bound the maximum number of panels (e.g., via panel area $\sim$1.8 m^2 per module), ensuring installable designs. Financial limits are imposed through a budget constraint (e.g., \$5k-\$50k range), including PV and storage costs with installation overhead. Key hardware cost assumptions and baseline tariff constants are summarized in Appendix A.4 (Table 12) for reference.

Operational constraints are encoded mathematically. Hourly energy balance ensures demand is met at every timestep (8760 hours), while battery dynamics enforce state-of-charge limits (10%–100%), efficiency ($\sim$97.5%), and charge/discharge rates (0.5 C). Export is restricted to available PV generation, and grid imports remain non-negative. By embedding these constraints directly in the MILP, every solution is guaranteed to be physically and economically feasible.

4.4.3 Agentic Decision Process

The SolarInvest Agent operates as a structured planner over a persistent conversation state, where the full dialogue history is always available (no truncation or sliding window). Each session maintains key inputs (e.g., location, EV count, occupants, budget, roof, tariff plan) alongside a pre-computed recommendation JSON. Rather than recomputing, the agent summarizes this structured output into a concise response (typically $\sim$1.5–2.5k characters) and uses it as the basis for subsequent reasoning. The decision process follows a perception–reasoning–action loop. Initial inputs are processed to generate two configurations: an optimal solution (efficiency-driven) and a recommended solution (bounded by budget and physical limits). For follow-ups, the agent incrementally updates responses using the growing conversation context (user $\sim$20–200 chars, assistant $\sim$100–500 chars per turn), without re-running underlying computations. This separation between tool execution (offline, deterministic) and interaction (LLM-driven, context-aware) ensures consistent decisions while allowing iterative refinement through dialogue.

4.5 Implementation Details (Engineering and Inference Optimization)

The system is implemented as a modular LLMOps pipeline with a backend-agnostic inference layer. Production inference uses the `grok-4.1-fast` model for balanced latency and reasoning capability, with support for larger models when deeper reasoning is required. Engineering optimizations include retry mechanisms, controlled decoding parameters, and strict token budgeting to ensure stable and predictable inference.

A lightweight retrieval layer injects relevant domain context into the prompt, while structured prompt construction and schema validation enforce consistency in outputs. A validation and repair mechanism ensures that outputs adhere to predefined formats, improving robustness in deployment.

5 Experiments and Results

5.1 Experimental Setup

5.2 Metrics

All error metrics are computed relative to the **Baseline as ground truth**, over N=20 scenarios each for the *unconstrained* and *budget-constrained* sizing objectives.

Panel count RMSE and MAE.

$$\mathrm{RMSE}_{\text{panels}} = \sqrt{\tfrac{1}{N}\sum_i \big(\hat{n}_i - n_i^{\mathrm{GT}}\big)^2}, \qquad \mathrm{MAE}_{\text{panels}} = \tfrac{1}{N}\sum_i \big|\hat{n}_i - n_i^{\mathrm{GT}}\big|.$$

Component	Baseline Task	Agentic Task 1	Agentic Task 2	Agentic Task 3
Task	Benchmark evaluation for Vanilla LLM Implementation	Benchmark evaluation for SolarInvestAgent	Grid evaluation 1	Grid evaluation 2
Description	Custom benchmark for stress testing for extreme scenarios	Custom benchmark for stress testing for extreme scenarios	Parameter sweep over uncorrelated features	Parameter sweep over correlated features
Constant parameters	N/A	N/A	Budget: $25,000 roof length: 8.0 m roof breadth: 6.25 m rate plan: TOU_DR panel brand: auto	EVs: 0 people: 2 daytime occupants: 1 rate plan: TOU_DR
Variable parameters	N/A	N/A	EVs: 0, 1, 2 people: 1, 2, 3 daytime occupants: 0, 1, 2	roof length: 5, 10, 15 m roof breadth: 5, 10, 15 m budget: $5,000, $10,000, $15,000 panel brand: REC Group, Canadian Solar, Aiko Solar
Locations	La Jolla, Oceanside	La Jolla, Oceanside	La Jolla, Oceanside, Coronado	La Jolla, Oceanside, Coronado
Test size	20	20	54	314
Models	llama3.1:8b	grok-4-1-fast-reasoning , grok-4-1-fast-non-reasoning		
Frameworks	Ollama	LangGraph		
Budget (calls/tokens/time)	Calls: 20 Tokens:∼55k Time: ∼25 min	Calls: 20 Tokens:∼150k Time: ∼15 min	Calls: 54 Tokens: ∼405k Time: ∼40 min	Calls: 243 Tokens: ∼1.8M Time: ∼3–4 hours
Data splits	N/A (inference only). Each scenario is independent; no training/validation split.			
Common tools	RAG KB, Open-Meteo, EIA load data, SDG&E tariffs			

Table 2: Experimental setup across benchmark and grid evaluation tasks.

Capex MAE and MAPE.

$$\text{MAE}_{\text{capex}} = \frac{1}{N}\sum_i \left|\hat{C}_i - C_i^{\text{GT}}\right|, \qquad \text{MAPE}_{\text{capex}} = \frac{1}{N}\sum_i \frac{|\hat{C}_i - C_i^{\text{GT}}|}{C_i^{\text{GT}}} \times 100\%.$$

Budget feasibility (higher is better). Per-scenario binary indicator: feasible if predicted CAPEX $\leq$ stated household budget ($\hat{C}_i \leq B_i$). A violation is a hard constraint failure: the proposed system cannot be built within budget.

5.3 Baselines

We evaluate four configurations:

- **Baseline (ground truth).** A deterministic solver-only pipeline that runs the full MILP optimizer directly in Python, without any LLM involvement. Given household inputs, it fetches real irradiance from the Open-Meteo API, models household load from EIA data, selects the optimal panel from a hardware catalog, and solves the constrained PV-BESS sizing problem using `HiGHS` via `highspy`. The output — panel count and gross CAPEX — serves as the reference for all error metrics.
- **Track A.** Same LLM with extended retrieval context (panel datasheets, tariff schedules, comparable installs), still without calling the MILP optimizer. [Instead of mentioning without calling the optimizer, we can tell it did not even correspond to direct constraints enforced]
- **Track B1.** SOLARINVESTAGENT (reasoning mode): `grok-4-1-fast-reasoning` orchestrates the MILP optimizer through step-by-step chain-of-thought reasoning before each tool call with an additional scenario-consistency validation after the optimizer returns results.
- **Track B2.** SOLARINVESTAGENT (non-reasoning mode): `grok-4-1-fast-non-reasoning` orchestrates the MILP optimizer through step-by-step chain-of-thought reasoning before each tool call with an additional scenario-consistency validation after the optimizer returns results.

Fairness. All tracks receive identical input features, tariff schedules, weather data, and panel cost assumptions. The only axes of variation are tool access and reasoning strategy.

5.4 Results

5.4.1 Unconstrained Sizing (Optimal)

Table 3 reports deviation from the Baseline reference under the unconstrained sizing objective, where the goal is pure lifecycle cost minimisation with no enforced budget cap.

Table 3: All tracks vs Baseline reference **Unconstrained sizing** (N=20). Lower RMSE/MAE/MAPE and higher feasibility are better. Baseline feasibility: 65.0% (13/20, 7 violations).

Track	RMSE (panels)	MAE (panels)	Capex MAE ($)	Capex MAPE (%)	Feasibility	Violations
Track A	277.35	186.85	60,585	10,468.5	25.0%	15/20
Track B1	**20.05**	**14.10**	**26,902**	**1,365.2**	**40.0%**	**12/20**
Track B2	20.05	14.10	26,902	1,365.2	40.0%	12/20

Note: Tracks B1 and B2 produce identical numeric outputs on this benchmark; the reasoning mode (Track B1) and non-reasoning mode (Track B2) differ in inference strategy but converge to the same MILP solution. High MAPE values arise because several Baseline capex entries are very small ($127-$760), amplifying percentage errors.

Track A vs Tracks B1/B2 Unconstrained. Table 4 directly compares Track A and Tracks B1/B2 against the Baseline reference on every metric. Tracks B1/B2 win on all dimensions: panel count RMSE is 13.8× lower (20.05 vs. 277.35), capex MAE is 2.3× lower ($26,902 vs. $60,585), and budget feasibility is higher (40% vs. 25%), demonstrating that MILP-grounded recommendations align substantially more closely with the deterministic Baseline.

Table 4: Track A vs Tracks B1/B2 head-to-head **Unconstrained sizing**. Error measured against Baseline reference. ✓marks the better value.

Metric	Track A	Track B1	Track B2	Better
Panels RMSE	277.35	**20.05**	**20.05**	B1/B2 ✓
Panels MAE	186.85	**14.10**	**14.10**	B1/B2 ✓
Capex MAE ($)	60,585	**26,902**	**26,902**	B1/B2 ✓
Capex MAPE (%)	10,468.5	**1,365.2**	**1,365.2**	B1/B2 ✓
Feasibility (%)	25.0	**40.0**	**40.0**	B1/B2 ✓
Violations	15/20	**12/20**	**12/20**	B1/B2 ✓

Note: B1 and B2 are numerically identical; both are shown for completeness.

5.4.2 Budget-Constrained Sizing (Recommended)

Table 5 reports deviations from the Baseline reference under the budget-constrained sizing objective, where all methods receive an explicit household budget to respect.

Table 5: All tracks vs Baseline reference **Budget-constrained sizing** (N=20). Baseline feasibility: 100.0% (20/20, 0 violations).

Track	RMSE (panels)	MAE (panels)	Capex MAE ($)	Capex MAPE (%)	Feasibility	Violations
Track A	86.44	63.70	18,220	1,738.2	85.0%	3/20
Track B1	**37.80**	**26.00**	**7,670**	**732.2**	75.0%	5/20
Track B2	37.80	26.00	7,670	732.2	75.0%	5/20

Note: Tracks B1 and B2 produce identical outputs. Track A has fewer budget violations (3 vs. 5) because its tendency to over-estimate panel counts happens to yield capex figures that stay within budget in more scenarios, not because of tighter constraint enforcement.

Track A vs Tracks B1/B2 Budget-constrained. The detailed head-to-head under the budget-constrained objective is provided in Appendix A.3 (Table 8). As noted in the table note, Track A's feasibility advantage stems from over-estimation: larger recommended panel counts incidentally correspond to capex figures that remain within budget in more scenarios, and this does not reflect superior constraint enforcement. As shown in Appendix Table 8, Tracks B1/B2 achieve lower error across all accuracy metrics: panel RMSE is 2.3× lower

(37.80 vs. 86.44), capex MAE is $2.4\times$ lower ($7,670 vs. $18,220), and capex MAPE is $2.4\times$ lower (732% vs. 1,738%).

5.4.3 Sensitivity and Robustness Analysis of Agentic PV Sizing

The sensitivity analysis across two complementary grid evaluations reveals consistent and interpretable behavior of the Agentic PV-sizing model under both structural and contextual variations. In the 243-point sweep, where roof dimensions, budget, and panel brand were varied, the number of recommended panels exhibited a strong positive dependence on available roof area (with near-linear scaling), while budget acted as a soft constraint—beyond a threshold, panel count saturated due to spatial limits rather than financial ones. CAPEX estimates showed a near-proportional relationship with panel count, with systematic offsets introduced by panel brand selection (premium brands increasing cost by approximately 15–25% for similar configurations).

In contrast, the 72-point sweep, which varied household demand parameters (number of EVs, occupants, and daytime presence) while keeping infrastructure fixed, demonstrated that panel recommendations scale primarily with energy demand drivers, particularly EV count and household size. Each additional EV resulted in a significant step increase in panel count, while increases in occupants had a more gradual but consistent effect. CAPEX in this setting closely tracked panel count, reinforcing that demand-side variation translates directly into system sizing rather than efficiency adjustments. Across both sweeps, location had minimal variance impact (due to similar solar profiles in the San Diego region), indicating that the model prioritizes structural constraints (roof area) and demand signals (EVs, occupants) over marginal environmental differences. Overall, the agent demonstrates stable monotonic behavior with respect to both supply-side (area, budget) and demand-side (load) inputs, with clear regime transitions where either spatial or financial constraints become dominant.

5.4.4 Overall Per-Track Summary

Table 6 consolidates average outputs and errors for all four tracks across both sizing objectives.

Table 6: Per-track summary over 20 scenarios. RMSE and Capex MAE are measured against the Baseline reference (zero by definition for Baseline).

Objective	Track	Avg panels	Avg capex ($)	Panels RMSE	Capex MAE ($)	Feas.	Viol.
Unconstrained (Optimal)	Baseline	33.9	21,811	0.00	0	65.0%	7/20
	Track A	220.7	81,602	277.35	60,585	25.0%	15/20
	Track B1	33.6	44,436	**20.05**	**26,902**	**40.0%**	**12/20**
	Track B2	33.6	44,436	20.05	26,902	40.0%	12/20
Budget-constrained (Recommended)	Baseline	32.4	17,394	0.00	0	**100.0%**	0/20
	Track A	84.1	34,812	86.44	18,220	85.0%	3/20
	Track B1	9.9	24,944	**37.80**	**7,670**	75.0%	5/20
	Track B2	9.9	24,944	37.80	7,670	75.0%	5/20

Note: Tracks B1 and B2 produce identical numeric outputs; the reasoning (B1) and non-reasoning (B2) inference modes converge to the same MILP solution, indicating that the chain-of-thought step does not alter the numeric recommendation on this benchmark.

1. *Tracks B1 and B2 are closest to the Baseline reference.* Across both objectives, Tracks B1 and B2 achieve the lowest panel count RMSE (20.05 unconstrained, 37.80 budget-constrained) and lowest capex MAE ($26,902 and $7,670 respectively), confirming that MILP-grounded recommendations align more closely with the deterministic Baseline than LLM-only approaches.
2. *Track A diverges substantially from the Baseline.* Without hard constraint enforcement, the LLM over-estimates system size dramatically: average panels jump from 33.9 (Baseline) to 220.7 (Track A) under the unconstrained objective, and from 32.4 to 84.1 under the budget-constrained objective. This inflates capex MAE to $60,585 and $18,220 respectively $2.3\times$ and $2.4\times$ worse than Tracks B1/B2.
3. *Tracks B1/B2 outperform Track A on all accuracy metrics.* In both objectives, Tracks B1 and B2 achieve lower RMSE, MAE, and MAPE than Track A. Track A records slightly fewer budget violations under the budget-constrained objective (3 vs. 5), but this is a coincidental consequence of its over-estimation pattern rather than improved constraint enforcement.
4. *Reasoning mode does not change numeric outputs.* Tracks B1 (reasoning) and B2 (non-reasoning) produce identical panel counts and capex on all 20 scenarios, indicating that the chain-of-thought step does not alter the MILP solution on this benchmark. The value of the reasoning mode may lie in explanation quality and traceability, which are not captured by these metrics.
5. *Budget feasibility is objective-dependent.* The Baseline achieves 100% feasibility under the budget-constrained objective by construction, but drops to 65% (7 violations) under the unconstrained objective

where no budget is explicitly provided. Tracks B1/B2 incur 5 violations under the budget-constrained objective, all concentrated in scenarios with budgets $\leq$\$9,000 where the MILP's minimum viable configuration (\$12,650) inherently exceeds the stated budget.

5.4.5 Example Trajectory

User: I live in La Jolla. Budget \$15,000, roof 3 m $\times$ 2.5 m, no EVs, 1 occupant, nobody home during the day. What solar system do you recommend?

Agent plan: (1) Fetch La Jolla irradiance via Open-Meteo. (2) Pull single-occupant EIA load profile. (3) Retrieve SDG&E TOU-DR tariff schedule. (4) Call `pv_bess_optimizer`(budget=\$15,000, roof=7.5 m^2, evs=0). (5) Validate JSON output against schema; repair if needed.

Tool call: `pv_bess_optimizer`({budget: 15000, roof_area: 7.5, evs: 0, people: 1, tariff: "TOU_DR"})

Tool output: {panels: 1, capex: \$14,198.80, annual_savings: \$892, payback_yr: 15.9, feasible: true}

Validation: Schema OK; Capex \$14,199 $\leq$ budget \$15,000; SoC bounds [10%-100%] satisfied; Power balance feasible over 8,760 h

Final output: "For your La Jolla home we recommend a **1-panel starter system with BESS** at a total installed cost of \$14,199 - within your \$15,000 budget. Under TOU-DR you can expect \$892/year in savings with a payback period of $\approx$16 years. Given your small roof (7.5 m^2) and low single-person load, one panel with battery storage captures most of the available arbitrage value. Adding a second panel would require a larger budget."

Figure 2: SolarInvestAgent (Track B) trajectory for scenario 6 (*tiny_roof_minimal_load*, La Jolla, Recommended case). The agent calls the MILP optimizer, validates all constraints, and produces a grounded natural-language recommendation. The Baseline (GT) output for the same scenario: 2 panels, capex \$12,903 — within budget but without constraint verification.

6 Conclusion

We presented SolarInvestAgent, a hybrid decision-support system that combines LLM-based orchestration with solver-backed optimization for residential PV–BESS sizing. Our results show that embedding a deterministic MILP core substantially improves numerical accuracy, constraint satisfaction, and financial consistency compared to LLM-only baselines. In particular, the agentic approach achieves significantly lower sizing and cost errors while avoiding the infeasible and over-estimated configurations commonly produced by direct LLM inference.

More broadly, this work highlights the importance of separating reasoning from computation in LLM systems: while LLMs are effective for synthesis and interaction, reliable decision-making in structured domains requires explicit optimization and verification. We believe this hybrid paradigm is a promising direction for building trustworthy AI systems in energy planning and other constraint-critical applications.

Table 7: P9 (Solar investment): traceability and scenario consistency

Model / Variant	NPV Err. (%)	Payback Err. (yr)	Traceable (%)	Scenario Cons. (%)	Safe Fail (%)	Tool Calls
Single-pass (no loop)	-	-	-	-	-	-
Agentic (verification-gated)	-	-	-	-	-	-
Baseline (deterministic model)	-	-	-	-	-	-

Note: Columns are examples. Teams may add sensitivity stability (variance across what-if perturbations) and auditability fields (e.g., all numbers tied to tool outputs).

A Additional Figures and Tables

A.1 Feature-engineering workflow

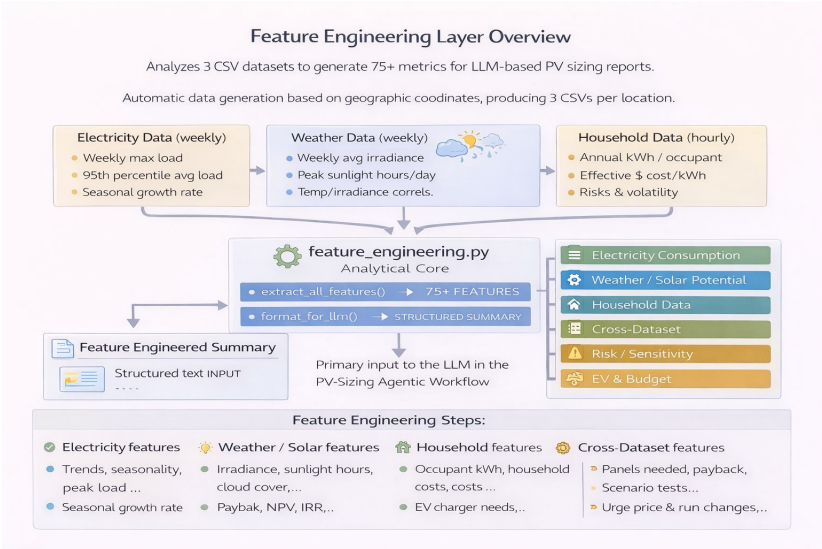

Figure 3: Feature engineering and aggregation pipeline used to transform raw weather/load signals into structured optimization-ready features.

A.2 Additional analysis plots

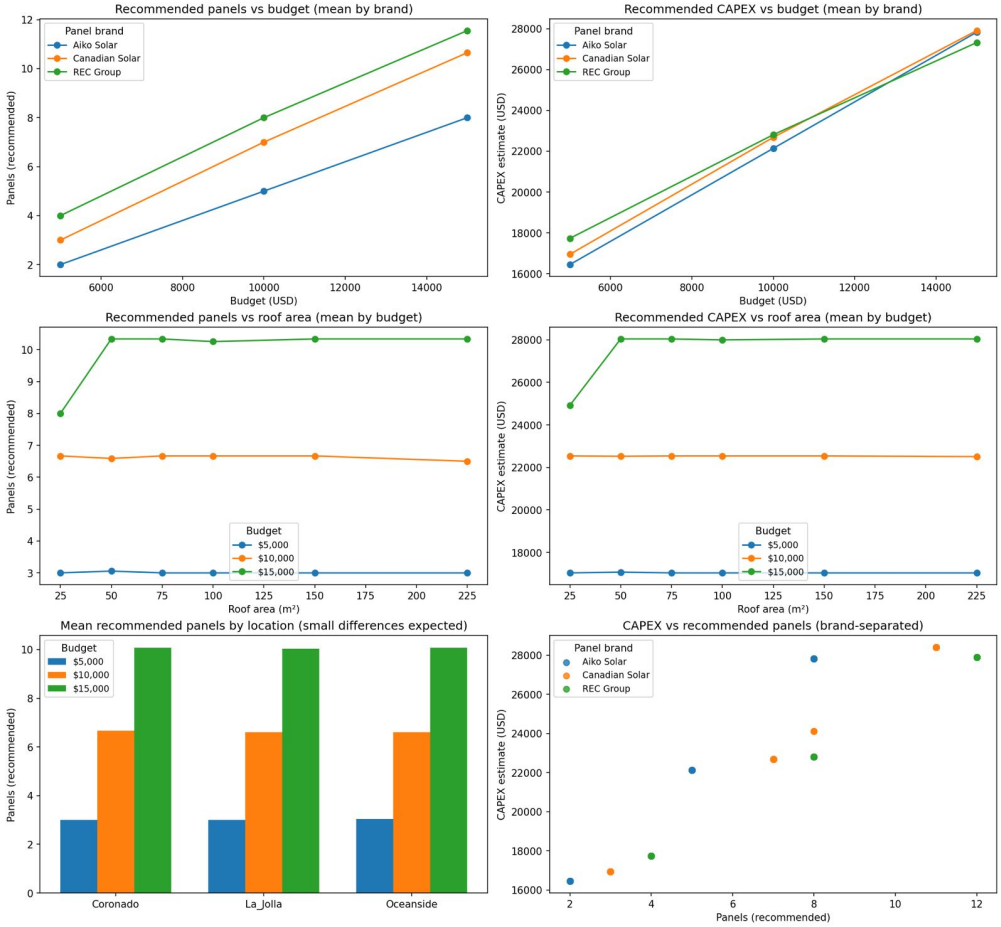

Figure 4: Analysis plot for case 243.

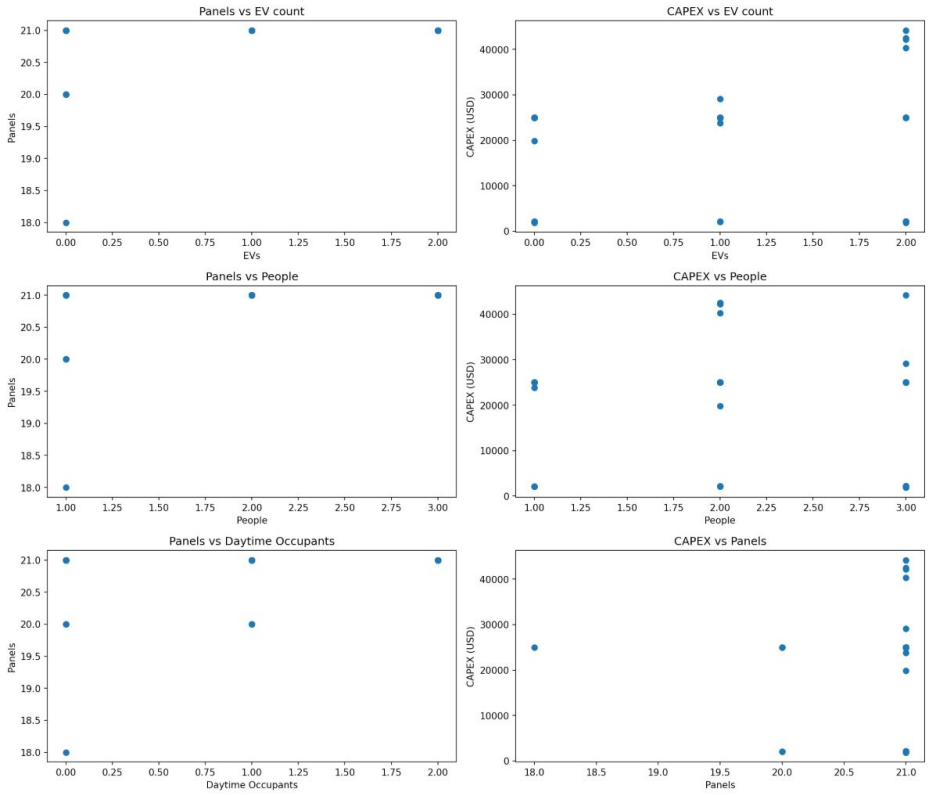

Figure 5: Analysis plot for case 72.

A.3 Budget-constrained head-to-head table

Table 8: Track A vs Tracks B1/B2 head-to-head **Budget-constrained sizing**. Error measured against Baseline reference. ✓marks the better value.

Metric	Track A	Track B1	Track B2	Better
Panels RMSE	86.44	**37.80**	**37.80**	B1/B2 ✓
Panels MAE	63.70	**26.00**	**26.00**	B1/B2 ✓
Capex MAE ($)	18,220	**7,670**	**7,670**	B1/B2 ✓
Capex MAPE (%)	1,738.2	**732.2**	**732.2**	B1/B2 ✓
Feasibility (%)	**85.0**	75.0	75.0	Track A ✓
Violations	**3/20**	5/20	5/20	Track A ✓

A.4 San Diego locations

Location	Latitude	Longitude
San Diego (Downtown)	32.7160	117.1611
Chula Vista	32.6399	117.1067
Oceanside	33.1959	117.3795
Escondido	33.1247	117.0808
Carlsbad	33.1581	117.3506
Vista	33.1936	117.2411
San Marcos	33.1434	117.1661
Encinitas	33.0391	117.2954
National City	32.6692	117.0890
Imperial Beach	32.5839	117.1131
El Cajon	32.7948	116.9628
La Mesa	32.7678	117.0230
Lemon Grove	32.7335	117.0337
Santee	32.8384	116.9739
Poway	32.9627	117.0362
Solana Beach	33.0006	117.2688
Alpine	32.8351	116.7664
Bonita	32.6653	117.0444
Fallbrook	33.3764	117.2511
Jamul	32.7266	116.8823
Coronado	32.6859	117.1831
Del Mar	32.9595	117.2653
Rancho Santa Fe	33.0167	117.2056
Camp Pendleton North	33.2817	117.3197
Eucalyptus Hills	32.8295	116.8201
La Jolla	32.8427	117.2578
Mira Mesa	32.9185	117.1382
Lakeside	32.8820	116.9017
Casa de Oro-Mt. Helix	32.7335	116.9964
Bostonia	32.8041	116.9470

Table 9: Locations used in benchmark and grid evaluations.

A.5 TOU tariff assumptions

Period	Buy ($/kWh)	Export ($/kWh)
Off-peak	0.32	0.08
Mid-peak	0.44	0.10
On-peak	0.59	0.13

Table 10: Representative TOU tariffs used for evaluation.

A.6 Hardware and cost assumptions

Panel brand	Efficiency (%)	Installed cost ($/W$_p$)	Power (W)
REC Group (Alpha Pure)	22.6	2.85	405
JA Solar (DeepBlue)	-	2.90	395
Trina Solar (Vertex S)	21.8	2.90	400
Canadian Solar (TOPHiKu7)	22.5	3.10	720
Silfab Solar (Prime)	22.1	3.15	440
Jinko Solar (Tiger Neo)	23.8	3.20	440
LONGi Solar (Hi-MO 6)	23.3	3.35	435
Maxeon Solar (Maxeon 7)	22.8	3.50	445
Aiko Solar (Neostar 2P)	24.3	3.75	485

Table 11: Panel assumptions used in scenario studies.

Cost component	Value
Inverter and BOS adder	$0.55/W$_p$
Installation labor adder	$0.40/W$_p$
Battery pack cost	$410/kWh
Annual O&M rate	1.0% of capex

Table 12: Cost constants used in optimization and financial analysis.

Table 13: Hardware Costs and Baseline Tariff Parameters for the San Diego Model

Parameter	Value	Source
Average PV Installed Cost	$2.85-$3.75/W$_p$ (avg. $3.14/W)	EnergySage [2025]
Average BESS Installed Cost	$7,000-$18,000/unit (10-13.5 kWh)	Industry data (2025)
TOU On-Peak Rate (Summer 2025)	$0.599/kWh	San Diego Gas & Electric [2025]
TOU Off-Peak Rate (Summer 2025)	$0.528/kWh	San Diego Gas & Electric [2025]
TOU Super Off-Peak (Summer 2025)	$0.450/kWh	San Diego Gas & Electric [2025]
Daily Fixed Charge (2025)	$0.402/day (Min. Bill)	San Diego Gas & Electric [2025]
NEM 3.0 Export Compensation	$0.05-$0.08/kWh	California Public Utilities Commission [2022]
Panel Temp. Coefficient (γ)	-0.004 %/°C	Jordan and Kurtz [2013]
Performance Ratio (PR)	0.75-0.85	IEA-PVPS (2020)
Annual Degradation Rate (d)	0.5%/year	Jordan and Kurtz [2013]
BESS Round-Trip Efficiency	90-95%	Manufacturer datasheets
Annual O&M Cost	$350-$500/year	NREL [2024]

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